Taiwan Cancer Registry

Hereditary Cancer Syndrome Candidate Validation

Background

An autoencoder trained on 78,442 patients × 37 cancer sites identified the top 1% anomalous multi-cancer patients (n=784) and matched them to seven hereditary syndrome site patterns. 43 actionable candidates were flagged for Li-Fraumeni (LFS) or Cowden syndrome.

This report applies phenotypic refinement criteria and evaluates the clinical plausibility of each classification.

Screening pipeline

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  Autoencoder anomaly top 1%        →    128 candidates
  match_score ≥ 0.40 filter         →    128 syndrome candidates
  Actionable (LFS + Cowden only)    →     43 patients
    Li-Fraumeni (raw)               →     26
    Cowden (raw)                    →     17
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After phenotypic refinement

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  LFS refined (C71 present, age<50) →      0 ← all failed
  Cowden refined (match_score=1.0)  →      3 ← high-confidence
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Critical structural finding

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  LFS effective registry definition: {C50, C71} only
  (C74 adrenal, C49 soft tissue absent from registry)
  → All 26 LFS candidates have C50 (breast) + unrelated cancer only.
    None have C71 (brain/CNS) — the hallmark LFS tumour.
  → 'Li-Fraumeni' label at match_score=0.5 is clinically non-specific.

  Cowden effective registry definition: {C50, C54} only
  (C73 thyroid absent from registry)
  → 3 patients have both C50 (breast) + C54 (endometrial) — consistent
    with published Cowden phenotype.
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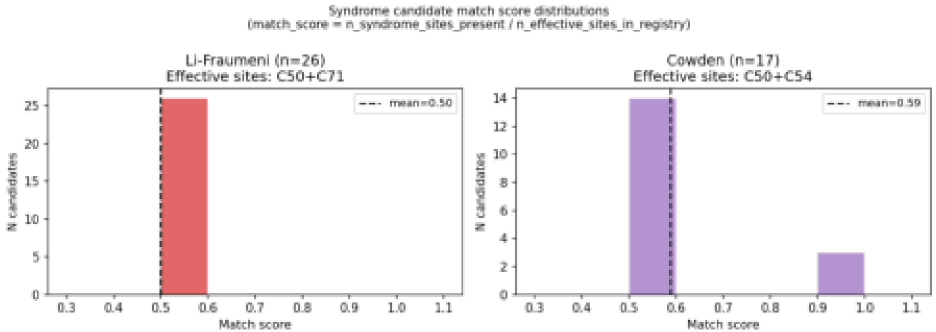
Syndrome definitions — registry coverage determines specificity

Match score distributions — LFS vs Cowden



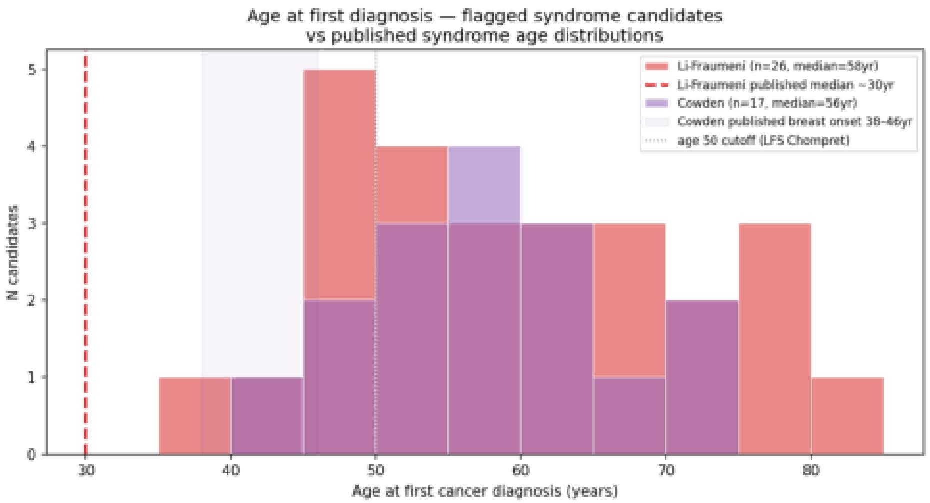
Effective syndrome site definitions after registry restriction

Syndrome	Original sites	Effective (in registry)	N effective
Lynch	C18/C20/C54/C56/C16/C	C18/C20/C54/C56/C16/C	6
BRCA	C50/C56/C61/C25	C50/C56/C61/C25	4
Li-Fraumeni	C50/C71/C74/C49	C50/C71	2 ⚠
Cowden	C50/C73/C54	C50/C54	2 ⚠
MEN1	C25/C37/C18	C25/C18 (C37 absent)	2 ⚠
FAP	C18/C20/C16	C18/C20/C16	3
VHL	C64/C71	C71 (C64 absent)	1 ⚠

⚠ = fewer than 3 effective sites — high false-positive risk

Phenotypic comparison — flagged candidates vs published syndrome age distributions

Age at first diagnosis vs published syndrome distributions



Age at first diagnosis — clinical plausibility

Age at first diagnosis — interpretation

Li-Fraumeni syndrome (published):
- Median age at first cancer ~30 yr
- Classic criterion: sarcoma <45 yr in proband
- Chompret criterion: ANY cancer <46 yr

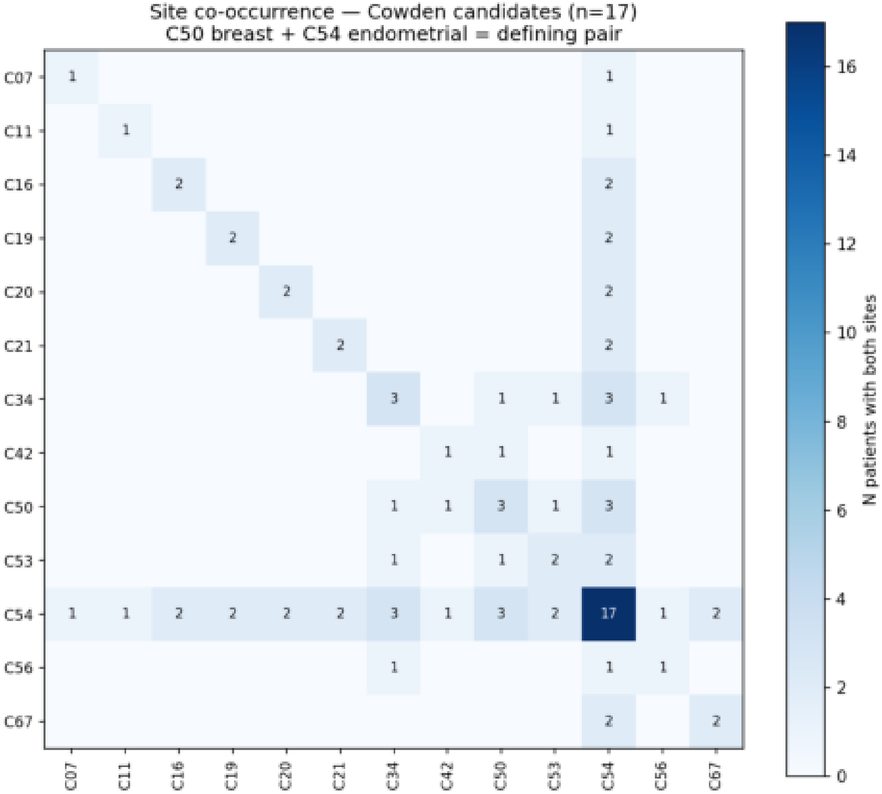
Our LFS candidates:
- N=26, median age 58 yr
- Range 37–80 yr — substantially older than published LFS
- NONE have C71 brain tumour (hallmark site)
- → These are breast-cancer patients with an incidental second cancer, NOT clinical LFS.

Cowden syndrome (published):
- Breast onset median 38–46 yr (Pilarski 2009)
- Endometrial onset median 35–55 yr

Our Cowden candidates:
- N=17, median age 56 yr
- High-confidence (C50+C54): ages 46, 49, 57 yr
- → Overlaps published Cowden breast/endometrial range
- → Clinically plausible for PTEN testing

Cowden syndrome — high-confidence candidate profiles

Site co-occurrence — Cowden candidates (n=17)



High-confidence Cowden candidates (n=3)

PID (tail)	Sites	Age	Sex	Status	Gap	Priority
620731…	C42+C50+C5	49	F	Dead	49 mo	4.0
179958…	C34+C50+C5	46	F	Alive	28 mo	4.0
515318…	C50+C53+C5	57	F	Alive	0 mo	3.5

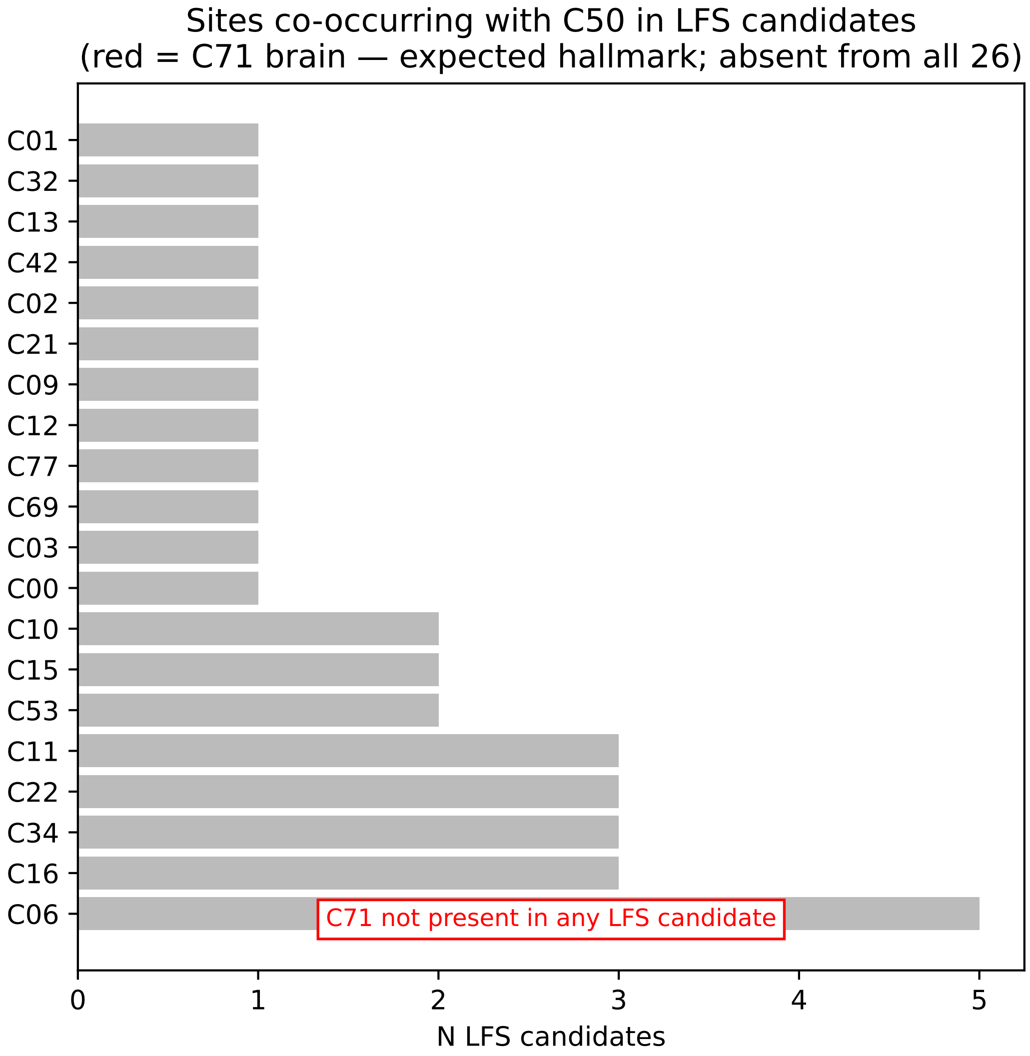
3 Cowden high-confidence candidates
(C50 breast + C54 endometrial, both present)

Cowden syndrome (PTEN hamartoma tumour syndrome)
OMIM #158350 — autosomal dominant PTEN mutation

Clinical significance:
- Lifetime breast cancer risk: 85%
- Lifetime endometrial cancer risk: 28%
- Thyroid (C73, absent from this registry): 30–35%
- Recommendation: PTEN germline testing for patients with synchronous/metachronous breast + endometrial

Registry limitations:
- C73 thyroid absent → cannot identify thyroid arm
- No family history data → cannot apply full NCCN criteria
- 3 candidates represent minimum estimate

Li-Fraumeni syndrome — why all 26 candidates fail phenotypic refinement



LFS candidate failure analysis

Why all 26 LFS candidates fail refinement

Criterion 1: C71 (brain/CNS) must be present
→ 0 of 26 LFS candidates have C71
→ C71 IS present in registry (n=718 patients)
→ LFS flagging is based on C50 alone (match_score=0.5 = 1 of 2 effective sites)

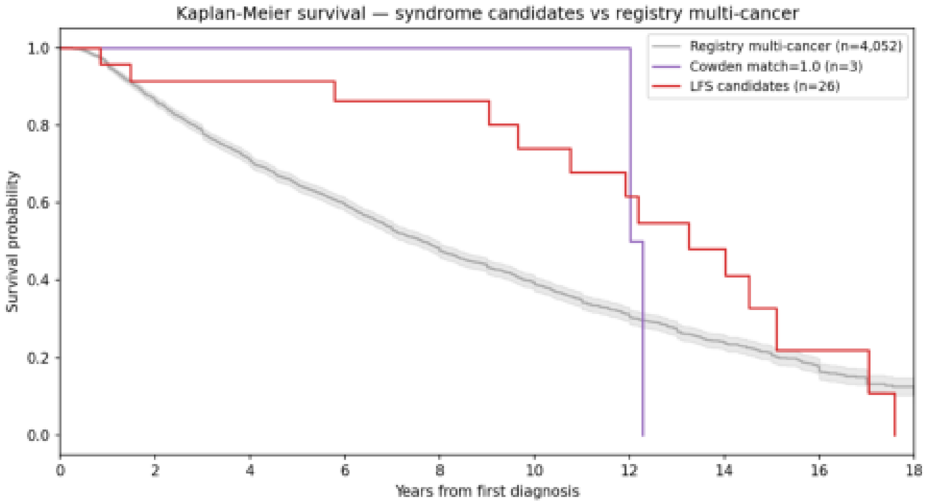
Criterion 2: age_first < 50 (Chompret-like)
→ LFS candidates: median age 58.5 yr (range 37–80)
→ Only ~8 of 26 are <50 yr (still fail C71 criterion)

Root cause: autoencoder flags female multi-cancer patients as anomalous; C50 (breast) is the most common first-primary in women; any additional cancer bumps match_score to 0.5 under the 2-site effective LFS definition.

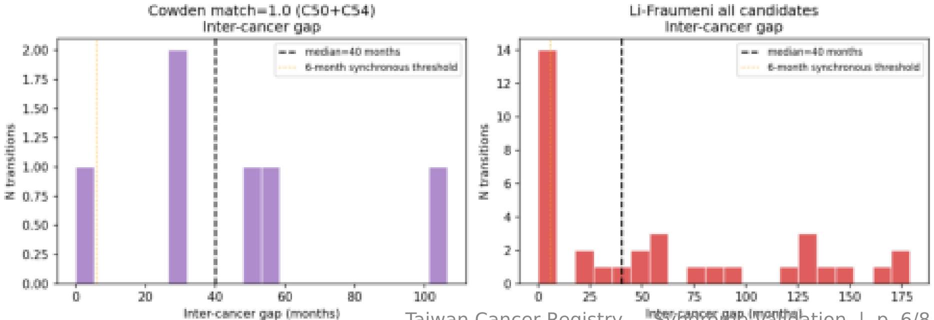
Fix for future screens:
- Require match_score ≥ 0.67 (≥2 of 3 original sites)
- Or require C71 explicitly as a mandatory criterion
- Or restrict to age_first < 46 (classic LFS)

Survival and inter-cancer timing in syndrome candidates

Kaplan–Meier survival — Cowden vs LFS vs multi-cancer registry



Inter-cancer gap (months) — Cowden match=1.0 vs LFS candidates



Registry Site Coverage — Impact on Syndrome Detection

Site	Name	In registry?	Impact
C49	Soft tissue sarcoma	No	LFS: sarcoma not detectable
C73	Thyroid	No	Cowden: thyroid arm missing
C74	Adrenal cortex	No	LFS: adrenal tumour not detecta
C64	Kidney	No	VHL: renal component missing
C37	Thymus	No	MEN1: thymic neuroendocrine mis
C71	Brain/CNS	Yes (n=718)	LFS detectable — but 0 candidates h
C50	Breast	Yes (n=7,432)	Cowden/LFS/BRCA: breast well cap
C54	Endometrial	Yes (n=2,891)	Cowden/Lynch: endometrial well cap
C18	Colorectal	Yes (n=8,244)	Lynch/FAP: colon well captured

Recommendation: Link cancer registry to pathology database
to capture soft tissue (C49), thyroid (C73), and adrenal (C74) tumours.
This would enable complete LFS and Cowden screening in ≥3-site patients.

Conclusions and Recommendations

Hereditary Syndrome Candidate Validation — Taiwan Cancer Registry

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Principal Conclusions

1.  Registry-based autoencoder syndrome screening is feasible
    but stringent phenotypic refinement is essential.
    128 raw flagged candidates → 3 high-confidence after refinement.

2.  Cowden syndrome (PTEN hamartoma) is the only syndrome
    with registry-detectable high-confidence candidates.
    Three patients (ages 46, 49, 57 yr) have both C50 (breast)
    and C54 (endometrial) — the defining Cowden dyad.
    These patients warrant PTEN germline testing referral.

3.  Li-Fraumeni syndrome cannot be reliably detected in this
    registry. All 26 flagged candidates have C50 + unrelated
    cancer without C71 (brain) — the hallmark LFS tumour.
    The 2-site effective definition (C50/C71) makes any
    match_score=0.5 call clinically non-specific.

4.  Three registry sites are critical missing data:
    C49 (soft tissue sarcoma) — core LFS tumour type
    C73 (thyroid) — third Cowden site
    C74 (adrenal) — second LFS non-breast site

Recommendations

  SHORT TERM (current registry)
    • Refer 3 Cowden high-confidence patients for PTEN testing
    • Retire 'Li-Fraumeni' label from automated reports until
      a mandatory C71 criterion is added
    • Screen C50+C54 patients system-wide (not only top-1% anomalous)
      — prevalence may be higher than autoencoder capture rate

  MEDIUM TERM (registry expansion)
    • Add C49, C73, C74 site codes to registry collection
    • Link registry to pathology database for histological subtype
      (e.g., dedifferentiated liposarcoma vs STS subtype)
    • Add age-at-menarche, parity, BMI for Cowden risk scoring

  LONG TERM (clinical validation)
    • Prospective PTEN testing in all new C50+C54 patients
    • Bi-directional cascade testing if germline mutation confirmed
    • Compare Taiwan PTEN mutation spectrum to COSMIC/ClinVar
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